

Financial Econometrics

Instrumental Variable Regression

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The government can use taxes on cigarettes to discourage smoking due to its negative externalities. The theory of optimal taxation would predict that the optimal tax rate is inversely related to the price elasticity. Hence, the greater the elasticity, the less quantity is affected by a given percentage tax, so the smaller is the change in consumption.

Unfortunately, price and quantity are jointly determined. In order to overcome the **simultaneous causality bias**, we need an instrument.

The *Cigarettes.csv* file provides a dataset allowing you to explore the aforementioned question. In specific, you may find following variables:

- *year* is fiscal year 1985 and 1995
- *state* is the U.S. state
- *tax* is cigarettes' specific tax
- *tax_sales* is general sales tax
- *avgprs* is natural log of average price per pack
- *lpackpc* is natural log of number of cigarettes packs per capita
- *income_capita* is natural log of income per capita
- All variables are already adjusted for inflation if needed

You are interested in estimating the elasticity of demand with respect to prices controlling for income per capita and state and year fixed effects.

$$lpackpc_{i,t} = \beta_i + \lambda_t + \beta_1 \times avgprs_{i,t} + \beta_2 \times income_capita_{i,t} + \varepsilon_{i,t}$$

In order to overcome the simultaneous causality bias, you use cigarettes' specific tax and general sales tax as instruments for cigarette price. Hence, the first stage regression takes following form.

$$avgprs_{i,t} = \pi_i + \delta_t + \pi_1 \times tax + \pi_2 \times tax_sales + \pi_3 \times income_capita_{i,t} + \varepsilon_{i,t}$$

Note that we exploit the panel set up and include state and year fixed effects into the model (also into the first stage regression).

- Task 1: Preparing the data
 - Load the data into R
 - Generate summary statistics using *summary()*
 - Declare panel data set
- Task 2: Use a *stargazer* table to report following five estimated models
 - Model 1: Naive OLS regression.
 - Model 2: First stage of IV regression. Test for instrument relevance using *linearHypothesis()*.
 - Model 3: Second stage regression using the fitted values from the previous model. Note that the obtained standard errors are not correct as they do not take into account that we use fitted values.
 - Model 4: Implement IV regression in one go using *plm()*.
 - Test for instrument exogeneity.

- Model 5: Redo Task 2.4 using only *tax_sales* as an instrument. The rationale is that cigarettes' specific tax *tax* is likely to be not exogenous.
- Task 3: Interpretation
 - What is the expected effect of increasing cigarette tax by 1% on demand?
 - Are the instruments relevant?
 - Are the instruments exogenous?
 - Is the naive specification subject to substantial simultaneous causality bias?

Make sure that you cluster standard errors by state [Use *coeftest()* and *vcovHC()*].

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#-----#
#           SOLUTION           #
#-----#

#-----#
# packages needed
require(plm)
require(lmtest)
require(sandwich)
require(car)
require(stargazer)

# clear environment
rm(list=ls(all=TRUE))

#-----#
# TASK 1

# Read data from the working directory.

setwd("C:/Users/s13163/Dropbox/FIE401/data/data_labs/")
CIG <- read.csv("Cigarettes.csv", header=TRUE, sep=",")

# Make summary statistics
summary(CIG)

##      state      year      tax      tax_sales
## Length:96      Min.   :1985  Min.   :16.73  Min.   : 0.0000
## Class :character 1st Qu.:1985  1st Qu.:26.94 1st Qu.: 0.8128
## Mode  :character Median :1990  Median :31.17 Median : 4.7901
##              Mean  :1990  Mean  :32.33 Mean  : 4.1269
##              3rd Qu.:1995  3rd Qu.:38.01 3rd Qu.: 5.9681
##              Max.   :1995  Max.   :64.96 Max.   :10.2642
##      lavgprs      lpckpc      lincome_capita
## Min.   :4.369      Min.   :3.897  Min.   :2.222
## 1st Qu.:4.559      1st Qu.:4.527  1st Qu.:2.514
## Median :4.640      Median :4.702  Median :2.615
## Mean   :4.673      Mean   :4.665  Mean   :2.614
## 3rd Qu.:4.771      3rd Qu.:4.816  3rd Qu.:2.716
## Max.   :5.063      Max.   :5.288  Max.   :3.043

# Declare panel data set
CIG<-pdata.frame(CIG,index=c("state","year"))
#-----#
# TASK 2 - Estimate five regression models

# Model 1: naive panel regression.
ols_fit <-plm(lpckpc ~ lavgprs+lincome_capita,
              data=CIG,model="within",effect="twoway")
cluster_se_ols <- coeftest(ols_fit,vcov=vcovHC(ols_fit,cluster="group",
                                              type="sss"))[,2]

# Model 2: first stage of IV regression with two instruments
stage1 <- plm(lavgprs ~ lincome_capita + tax + tax_sales,
              data=CIG,model="within",effect="twoway")
cluster_se_stage1 <- coeftest(stage1, vcov=vcovHC(stage1,cluster="group",
                                              type="sss"))[,2]

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# extract fitted value
CIG$lavgprs_hat <- fitted(stage1)

# Check instrument relevance
myH0 <- c("tax=0", "tax_sales=0")
linearHypothesis(stage1, myH0, test="F", vcov=vcovHC(stage1, cluster="group",
                                                    type="sss"))

##
## Linear hypothesis test:
## tax = 0
## tax_sales = 0
##
## Model 1: restricted model
## Model 2: lavgprs ~ lincome_capita + tax + tax_sales
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F    Pr(>F)
## 1      46
## 2      44  2 92.665 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#Conclusion: instruments are relevant

# Model 3: second stage of IV regression with two instruments
stage2 <- plm(lpackpc ~ lavgprs_hat+lincome_capita,
              data=CIG, model="within", effect="twoway")
cluster_se_stage2 <- coeftest(stage2, vcov=vcovHC(stage2, cluster="group",
                                                type="sss"))[,2]

# Model 4: IV regression in one go with two instruments
iv_fit_2I <- plm(lpackpc ~ lavgprs+lincome_capita |
                tax + tax_sales + lincome_capita,
                data=CIG, model="within", effect="twoway")
cluster_se_iv_2I <- coeftest(iv_fit_2I, vcov=vcovHC(iv_fit_2I, cluster="group",
                                                    type="sss"))[,2]

#Check instruments exogeneity
CIG$res<-residuals(iv_fit_2I);
exog<-plm(res~tax+tax_sales+lincome_capita,
          model="within", effect="twoway", data=CIG);
myH0 <- c("tax=0", "tax_sales=0")
Fstat<-linearHypothesis(exog, myH0, test="F")$F[2]
print(Fstat)

## [1] 2.465997
testval<-2*Fstat;
print(testval)

## [1] 4.931993
pval<-1-pchisq(testval,1);
print(pval)

## [1] 0.02636389
#We reject the H0: all instruments are exogenous

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#Tax specific for cigarettes is more likely to be endogenous, than general sales tax
#Use only general sales tax as instrument

# Model 5: IV regression in one go with one instruments
iv_fit_1I<- plm(lpackpc ~ lavgprs+lincome_capita |
               tax_sales + lincome_capita,
               data=CIG,model="within",effect="twoway")
cluster_se_iv_1I <- coeftest(iv_fit_1I,vcov=vcovHC(iv_fit_1I,cluster="group",
                                                type="sss"))[,2]

# Check instrument relevance

# Run first stage of IV regression with one instrument
stage1_1I <- plm(lavgprs ~ lincome_capita + tax_sales,
                 data=CIG,model="within",effect="twoway")

myH0 <- c("tax_sales=0")
linearHypothesis(stage1_1I, myH0,test="F",
                 vcov=vcovHC(stage1_1I,cluster="group",
                             type="sss"))

##
## Linear hypothesis test:
## tax_sales = 0
##
## Model 1: restricted model
## Model 2: lavgprs ~ lincome_capita + tax_sales
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F    Pr(>F)
## 1      46
## 2      45  1 34.8 4.406e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#Conclusion: instrument is relevant

# Check exogeniety: not possible since we have only one instrument

# stargazer output
stargazer(list(ols_fit, stage1, stage2, iv_fit_2I, iv_fit_1I),
          se = list(cluster_se_ols, cluster_se_stage1, cluster_se_stage2, cluster_se_iv_2I, cluster_se_iv_1I),
          type="text", keep.stat=c("n"), report=('vc*t'))

##
## =====
##                               Dependent variable:
##                               -----
##               lpackpc    lavgprs               lpackpc
##               (1)       (2)       (3)       (4)       (5)
## -----
## lavgprs      -1.056***
##               t = -6.699
##
## lavgprs_hat               -1.202***
##               t = -6.660
##

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## lincome_capita    0.497      -0.029      0.462      0.462      0.526
##                  t = 1.563  t = -0.239 t = 1.376  t = 1.518  t = 1.575
##
## tax                0.008***
##                  t = 8.742
##
## tax_sales          0.013***
##                  t = 4.382
##
## -----
## Observations      96          96          96          96          96
## =====
## Note:                                *p<0.1; **p<0.05; ***p<0.01

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